Name: ____________________ Partner: ____________________ Period: ________

The Tangent Function — Tiered Practice

The tangent function connects the unit circle to slope and to periodic asymptotic behavior. Choose **one tier** below based on your readiness level. Everyone should be prepared to explain their reasoning to the class.

Tier R — Remediate

Context: The unit circle values below are given. Use them to compute $\tan\theta = \sin\theta / \cos\theta$ at each angle.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
0	0	1	________
$\pi/6$	$1/2$	$\sqrt{3}/2$	________
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	________
$\pi/3$	$\sqrt{3}/2$	$1/2$	________
$\pi/2$	1	0	________
$3\pi/4$	$\sqrt{2}/2$	$-\sqrt{2}/2$	________
π	0	-1	________

1. At which angles above is $\tan\theta$ *undefined*? Why? ____________________

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2. The tangent function has vertical asymptotes at $\theta = \pi/2 + k\pi$ for integer k. List the three asymptotes closest to $\theta = 0$ (one negative, one positive, one further positive). ____________________

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3. Between $\theta = 0$ and $\theta = \pi/2$, is $\tan\theta$ increasing or decreasing? How do you know from the table? ________

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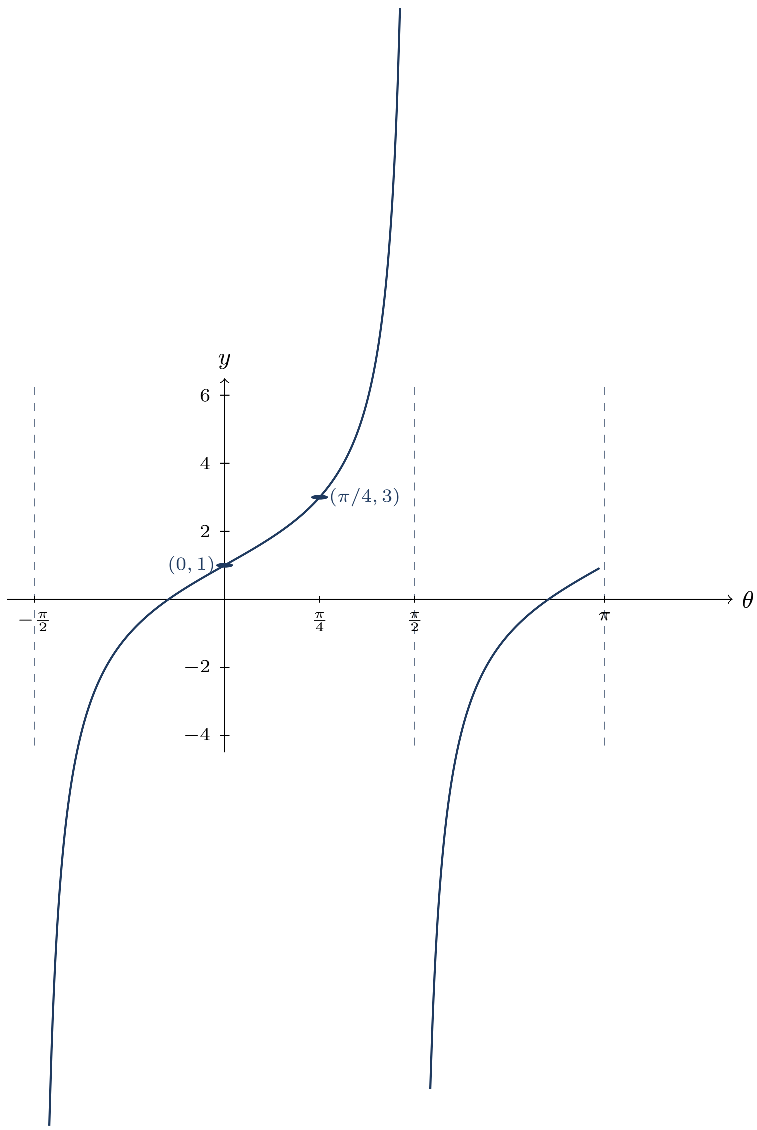
Tier A — Approaching Proficiency

Context: The graph below shows one and a half periods of a tangent function $f(\theta) = a\tan(\theta) + d$.



1. What is the period of f? ______

2. List all vertical asymptotes visible in the graph. ______

3. On the interval $(-\pi/2, \pi/2)$, is f concave up or concave down for $\theta < 0$? For $\theta > 0$? ______

______

4. The function passes through $(0, 1)$ and $(\pi/4, 3)$. Use these points to find the value of a and d in $f(\theta) = a\tan(\theta) + d$. Show your work. ______

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Tier E — Extension

Context: Analyze $g(\theta) = 2\tan\left(\frac{\theta}{2} - \frac{\pi}{4}\right) + 1$.

1. Rewrite g in the form $a\tan(b(\theta + c)) + d$ by factoring out $b = 1/2$ from the argument. Identify a, b, c, and d.

2. State the period of g. Show the computation.

3. Find all vertical asymptotes of g. (Set $\frac{1}{2}(\theta - \frac{\pi}{2}) = \frac{\pi}{2} + k\pi$ and solve for θ.)

4. Compute $g(0)$ and $g\left(\frac{\pi}{2}\right)$. Show your work.

5. Describe in words how the graph of g is related to the parent graph $y = \tan\theta$, addressing each transformation parameter.